

# FUNCTIONAL REQUIREMENTS DOCUMENT

## No-Show Reduction Initiative

SalonBookr — SalonOS Platform

|                         |                                                                     |
|-------------------------|---------------------------------------------------------------------|
| <b>Document Type</b>    | Functional Requirements Document (FRD)                              |
| <b>Product</b>          | SalonBookr (SalonOS B2B SaaS Platform)                              |
| <b>Initiative</b>       | Appointment No-Show Reduction                                       |
| <b>Owner</b>            | Product Team                                                        |
| <b>Status</b>           | Draft for Review                                                    |
| <b>Version</b>          | 2.0 — see Revision History below                                    |
| <b>Related Document</b> | Business Requirements Document (BRD) — No-Show Reduction Initiative |
| <b>Integrations</b>     | WhatsApp Business API, Razorpay Payment Gateway                     |

### Revision History

| Version | Change          | Summary                                                                                                                                                                  |
|---------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | Initial release | Reminder System, Waitlist Management (slot recovery), and mandatory Pre-Payment Collection                                                                               |
| 2.0     | Scope revision  | Feature 2 rewritten as Live Delay Tracker; Feature 3 rewritten as fully elective with discount and Priority Service logic; cross-feature rules and NFRs updated to match |

## Table of Contents

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|                                              |    |
|----------------------------------------------|----|
| Table of Contents                            | 2  |
| 1. Purpose and Document Scope                | 3  |
| 2. Actors and System Components              | 3  |
| 3. Feature 1 — Appointment Reminder System   | 3  |
| 3.1 Objective                                | 3  |
| 3.2 Reminder Schedule                        | 3  |
| 3.3 Functional Requirements                  | 4  |
| 3.4 Message Field Specification              | 4  |
| 3.5 Process Flow                             | 5  |
| 3.6 Acceptance Criteria                      | 5  |
| 4. Feature 2 — Live Delay Tracker            | 6  |
| 4.1 Objective                                | 6  |
| 4.2 Functional Requirements                  | 6  |
| 4.3 Delay Notification Cadence               | 6  |
| 4.4 Message Field Specification              | 7  |
| 4.5 Process Flow                             | 7  |
| 4.6 State Transitions — Delay Status         | 7  |
| 4.7 Acceptance Criteria                      | 8  |
| 5. Feature 3 — Pre-Payment System (Optional) | 9  |
| 5.1 Objective                                | 9  |
| 5.2 Functional Requirements                  | 9  |
| 5.3 Pre-Payment Discount Configuration       | 9  |
| 5.4 Priority Service Behavior                | 10 |
| 5.5 Payment Status Values                    | 10 |
| 5.6 Process Flow                             | 10 |
| 5.7 Acceptance Criteria                      | 11 |
| 6. Cross-Feature Business Rules              | 12 |
| 7. Non-Functional Requirements               | 12 |
| 8. Out of Scope (Functional)                 | 12 |
| 9. Open Questions                            | 13 |

## 1. Purpose and Document Scope

This Functional Requirements Document (FRD) translates the business requirements defined in the No-Show Reduction Initiative BRD into detailed system behavior. It specifies functional requirements, process flows, message and field specifications, business rules, state transitions, and acceptance criteria for engineering, QA, and implementation teams.

This document covers three functional modules: the Appointment Reminder System, the Live Delay Tracker, and the optional Pre-Payment System. It assumes the reader has reviewed the companion BRD for business context, objectives, and success metrics.

## 2. Actors and System Components

| Actor / System               | Role                                                                                                 |
|------------------------------|------------------------------------------------------------------------------------------------------|
| Customer                     | End consumer who books, receives reminders and delay updates, and may elect to pre-pay via WhatsApp  |
| Salon Owner / Staff          | Configures reminder settings, delay cadence, and pre-payment rules; manages the appointment calendar |
| SalonBookr Scheduling Engine | Core system managing appointment state, triggers, calendar data, and real-time delay calculation     |
| WhatsApp Business API        | External channel for confirmation, reminder, and delay status messages                               |
| Razorpay                     | External payment gateway for optional pre-payment collection                                         |
| CRM                          | Stores customer contact details and booking history                                                  |
| Analytics Engine             | Tracks and reports KPIs defined in the BRD                                                           |

## 3. Feature 1 — Appointment Reminder System

### 3.1 Objective

Reduce no-shows caused by customers forgetting their appointments by sending a structured sequence of WhatsApp messages from the time of booking through the appointment window.

### 3.2 Reminder Schedule

| Trigger Point               | Message Type         | Default State |
|-----------------------------|----------------------|---------------|
| Immediately after booking   | Booking Confirmation | Always on     |
| 24 hours before appointment | Reminder 1           | Always on     |
| 2 hours before appointment  | Reminder 2           | Always on     |

|                               |                       |                                    |
|-------------------------------|-----------------------|------------------------------------|
| 1 hour before appointment     | Reminder 3            | Always on                          |
| 30 minutes before appointment | Reminder 4 (Optional) | Off by default — salon must opt in |

Note: the 30-minute reminder is configurable per salon. All other reminders in the sequence are mandatory and cannot be disabled, since they are core to the no-show reduction objective defined in the BRD.

### 3.3 Functional Requirements

**FR-1.1:** The system shall send a booking confirmation message via WhatsApp immediately upon successful appointment creation (within 60 seconds of booking).

**FR-1.2:** The system shall trigger Reminder 1 exactly 24 hours before the scheduled appointment time.

**FR-1.3:** The system shall trigger Reminder 2 exactly 2 hours before the scheduled appointment time.

**FR-1.4:** The system shall trigger Reminder 3 exactly 1 hour before the scheduled appointment time.

**FR-1.5:** The system shall allow salon owners to enable an optional Reminder 4, sent 30 minutes before the scheduled appointment time, from the salon settings panel.

**FR-1.6:** If an appointment is booked less than 24 hours before its scheduled time, the system shall skip reminders whose trigger window has already passed and send only the remaining applicable reminders.

**FR-1.7:** If a customer reschedules an appointment, the system shall cancel all pending reminders tied to the original appointment time and schedule new reminders against the updated time.

**FR-1.8:** If a customer cancels an appointment, the system shall cancel all pending reminders for that appointment immediately.

**FR-1.9:** Each reminder message shall include: customer name, salon name, service booked, appointment date, appointment time, a reschedule action, and a cancel action.

**FR-1.10:** If a WhatsApp message fails to deliver, the system shall retry delivery up to 2 times over 10 minutes, then log the failure for analytics tracking.

**FR-1.11:** Where a Reminder System trigger coincides with a Live Delay Tracker trigger (Feature 2) at the same point in time, the two messages shall be merged into a single WhatsApp message (see Section 6, Cross-Feature Business Rules).

### 3.4 Message Field Specification

| Field            | Description                                        |
|------------------|----------------------------------------------------|
| Customer Name    | Pulled from CRM record associated with the booking |
| Salon Name       | Name of the salon as configured in SalonBookr      |
| Service Booked   | Service(s) selected at time of booking             |
| Appointment Date | Format: DD MMM YYYY                                |

|                   |                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------|
| Appointment Time  | Format: HH:MM AM/PM, salon's local time zone                                                   |
| Reschedule Action | Deep link to SalonBookr reschedule flow                                                        |
| Cancel Action     | Inline WhatsApp button that cancels the appointment and releases the slot back to the calendar |

### 3.5 Process Flow

**Step 1** — Customer completes a booking through call-based or online flow.

**Step 2** — Scheduling Engine creates the appointment record and computes reminder trigger timestamps (24h, 2h, 1h, and 30min if enabled).

**Step 3** — Scheduling Engine immediately sends the Booking Confirmation message via WhatsApp Business API.

**Step 4** — At each trigger timestamp, the Scheduling Engine queues the corresponding reminder for delivery, merging with any coinciding Live Delay Tracker update (see Section 6).

**Step 5** — Customer may tap Reschedule or Cancel directly from any reminder message.

**Step 6** — If Cancel is selected, the appointment status changes to Cancelled, remaining reminders are cleared, and the slot is released back to the salon's calendar.

**Step 7** — If Reschedule is selected, the customer is routed to select a new slot; reminders are recalculated against the new time.

**Step 8** — If no action is taken, the appointment proceeds to its scheduled time and the salon marks it as Attended or No-Show in the calendar.

### 3.6 Acceptance Criteria

- Given a new booking, when the booking is confirmed, then a confirmation message is sent within 60 seconds.
- Given an appointment scheduled 24+ hours out, when each trigger window is reached, then the corresponding reminder is sent.
- Given a salon has not enabled the 30-minute reminder, when the 30-minute window is reached, then no message is sent.
- Given a customer cancels via a reminder message, when the cancellation is processed, then all remaining reminders for that appointment are removed from the queue.
- Given an appointment is booked less than 24 hours before its scheduled time, when the booking is confirmed, then only reminders whose trigger window has not yet passed are scheduled.

## 4. Feature 2 — Live Delay Tracker

*Note: This module was originally scoped as "Waitlist Management" (slot recovery for cancellations). It has been redefined to address running delays during the appointment day instead.*

### 4.1 Objective

Reduce customer frustration and last-minute cancellations caused by salons running behind schedule, by proactively communicating real-time delay status to customers and offering a reschedule option when delays become significant.

### 4.2 Functional Requirements

**FR-2.1:** The Scheduling Engine shall calculate a running delay estimate for each upcoming appointment, based on the actual real-time progress of earlier appointments for the same staff member that day.

**FR-2.2:** Salons shall be able to enable or disable the Live Delay Tracker, and configure its notification cadence and reschedule threshold, from SalonOS settings.

**FR-2.3:** Starting 2 hours before a customer's scheduled appointment time, the system shall send a delay status update via WhatsApp every 30 minutes.

**FR-2.4:** Starting 1 hour before a customer's scheduled appointment time, the system shall increase the delay update frequency to every 15 minutes.

**FR-2.5:** Each delay update shall state the current estimated delay in plain language (e.g., "No delay", "Running approximately 15 minutes late", "Running approximately 30 minutes late").

**FR-2.6:** If the cumulative estimated delay exceeds 60 minutes past the customer's original scheduled appointment time, the system shall include a reschedule option in that and all subsequent updates.

**FR-2.7:** If the customer accepts the reschedule offer, the system shall route them to select a new slot; the original slot shall be released back to the salon's calendar.

**FR-2.8:** If the customer declines or does not respond to the reschedule offer, the system shall continue sending delay updates at the configured cadence until the customer is marked as seen.

**FR-2.9:** If an appointment is booked less than 2 hours before its scheduled time, the system shall begin delay tracking at whichever cadence is applicable at the time of booking.

**FR-2.10:** Delay tracking for an appointment shall stop once the salon marks the customer as Attended, Cancelled, or Rescheduled.

**FR-2.11:** Where a Reminder System trigger (Feature 1) coincides with a Delay Tracker trigger at the same point in time, the two shall be merged into a single WhatsApp message (see Section 6, Cross-Feature Business Rules).

### 4.3 Delay Notification Cadence

| Window                               | Update Frequency | Status    |
|--------------------------------------|------------------|-----------|
| 2 hours to 1 hour before appointment | Every 30 minutes | Always on |

|                                                 |                                           |                  |
|-------------------------------------------------|-------------------------------------------|------------------|
| 1 hour before appointment to appointment time   | Every 15 minutes                          | <b>Always on</b> |
| Cumulative delay exceeds 60 minutes (any point) | Reschedule option appended to next update | <b>Always on</b> |

Note: cadence and the 60-minute reschedule threshold are configurable per salon (FR-2.2); the values above reflect platform defaults.

## 4.4 Message Field Specification

| Field                   | Description                                                  |
|-------------------------|--------------------------------------------------------------|
| Customer Name           | Pulled from CRM record associated with the booking           |
| Salon Name              | Name of the salon as configured in SalonBookr                |
| Service Booked          | Service(s) selected at time of booking                       |
| Original Scheduled Time | The customer's original booked appointment time              |
| Current Estimated Delay | "No delay" / "~15 min" / "~30 min" / "~45 min" / "60+ min"   |
| Revised Estimated Time  | Original scheduled time plus current estimated delay         |
| Reschedule Action       | Shown only once cumulative delay exceeds 60 minutes (FR-2.6) |

## 4.5 Process Flow

**Step 1** — Customer's appointment is confirmed (with or without pre-payment).

**Step 2** — At the 2-hour mark before the scheduled time, the Scheduling Engine checks the salon's real-time schedule status for the relevant staff member.

**Step 3** — If the Live Delay Tracker is enabled, the system sends a delay status update via WhatsApp, merged with the Feature 1 reminder due at the same time.

**Step 4** — The system re-checks and sends updates every 30 minutes until the 1-hour mark.

**Step 5** — From the 1-hour mark onward, the system re-checks and sends updates every 15 minutes until the customer is seen.

**Step 6** — If, at any check, cumulative delay exceeds 60 minutes, the update includes a Reschedule action.

**Step 7** — If the customer taps Reschedule, they are routed to select a new slot; the original slot is released back to the calendar.

**Step 8** — If the customer takes no action, updates continue at the configured cadence until the salon marks the appointment as Attended.

## 4.6 State Transitions — Delay Status

| State                  | Trigger                                 | Next State                                  |
|------------------------|-----------------------------------------|---------------------------------------------|
| On Schedule (No Delay) | Real-time check shows salon on schedule | Minor Delay, if a delay begins accumulating |

|                            |                                               |                                                                     |
|----------------------------|-----------------------------------------------|---------------------------------------------------------------------|
| Minor Delay (1–29 min)     | Cumulative delay enters this band             | Moderate Delay, or back to On Schedule if recovered                 |
| Moderate Delay (30–59 min) | Cumulative delay enters this band             | Major Delay, or back to Minor/On Schedule if recovered              |
| Major Delay (60+ min)      | Cumulative delay crosses the 60-min threshold | Rescheduled (accepted) or remains Major Delay (declined) until seen |
| Resolved                   | Customer is marked as Attended                | Closed — delay tracking stops                                       |

## 4.7 Acceptance Criteria

- Given an appointment is 2 hours out, when the 2-hour mark is reached, then a delay status update is sent reflecting the salon's current schedule status.
- Given delay updates are being sent every 30 minutes, when the 1-hour mark is reached, then the update frequency increases to every 15 minutes.
- Given cumulative delay exceeds 60 minutes, when the next update is generated, then a reschedule option is included.
- Given a customer accepts a reschedule offer, when they select a new slot, then the original slot is released back into the salon's calendar.
- Given a customer declines or does not respond to a reschedule offer, when the next update is due, then delay status updates continue at the configured cadence.
- Given an appointment is booked less than 2 hours before its scheduled time, when the booking is confirmed, then delay tracking begins immediately at the cadence applicable to the remaining time window.
- Given an appointment is marked Attended by the salon, when the next scheduled check would occur, then no further delay updates are sent for that appointment.
- Given a salon's schedule recovers before the customer's appointment, when the next check runs, then the delay status returns to "No delay."

## 5. Feature 3 — Pre-Payment System (Optional)

### 5.1 Objective

Increase customer commitment and reduce casual, low-intent bookings by rewarding customers who voluntarily pre-pay with a percentage discount and priority service. Pre-payment is never required to book an appointment.

### 5.2 Functional Requirements

**FR-3.1:** Salons shall be able to enable Pre-Payment as an elective option for customers at the time of booking, for some or all services.

**FR-3.2:** Customers shall be able to choose, at the time of booking, whether to pre-pay (in full or in part, per salon configuration) or proceed with standard no-payment booking.

**FR-3.3:** Payment processing shall be handled exclusively through Razorpay.

**FR-3.4:** If a customer chooses to pre-pay and payment succeeds, the system shall confirm the appointment at the discounted total, tag the booking as Priority Service, store the payment record, and trigger the Reminder System (Feature 1).

**FR-3.5:** If a customer chooses to pre-pay and payment fails, the system shall not create an appointment record under the pre-payment path, and shall prompt the customer to retry payment or proceed with standard no-payment booking instead.

**FR-3.6:** All payment records (amount, discount applied, status, transaction ID, timestamp) shall be stored within SalonBookr and linked to the corresponding appointment.

**FR-3.7:** Customers shall receive a payment receipt via WhatsApp upon successful transaction.

**FR-3.8:** The system shall hold a payment-pending slot for a configurable window (default: 10 minutes) to prevent double-booking while payment is in progress; if payment is not completed within this window, the slot is released.

**FR-3.9:** Salon owners shall be able to configure a pre-payment discount percentage per service, applied to the total bill when a customer completes pre-payment.

**FR-3.10:** The system shall display the original price, discount percentage, discount amount, and final payable amount to the customer before payment is made.

**FR-3.11:** Customers who successfully complete pre-payment shall be tagged Priority Service for that appointment in the Scheduling Engine.

**FR-3.12:** The Scheduling Engine shall prioritize on-time service for Priority Service customers ahead of non-prepaid customers wherever staff scheduling allows.

**FR-3.13:** Priority Service customers shall receive a modified Live Delay Tracker experience: a single reassurance message ("Priority Service — minimal delay expected") in place of the standard escalating cadence, unless an actual delay materializes for that customer, in which case standard Feature 2 updates resume.

### 5.3 Pre-Payment Discount Configuration

| Setting             | Description                                            |
|---------------------|--------------------------------------------------------|
| Discount Percentage | Configurable per service (e.g., 5%–20%); salon-defined |

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| Applies To             | Full pre-payment only, or also partial pre-payment, per salon configuration |
| Display Point          | Shown at checkout before payment is made (FR-3.10)                          |
| Refund on Cancellation | Out of scope for this release — see Section 8, Out of Scope                 |

## 5.4 Priority Service Behavior

Pre-paid customers are tagged Priority Service in the Scheduling Engine. This status interacts directly with the Live Delay Tracker (Feature 2):

- Priority Service customers are scheduled with buffer protection where staffing allows, reducing the likelihood they inherit delay from earlier appointments.
- Priority Service customers receive a single reassurance message instead of the standard escalating delay cadence, unless an actual delay occurs (FR-3.13).
- If a Priority Service customer's appointment is nonetheless delayed beyond 60 minutes, the standard Feature 2 reschedule flow applies in full, and any discount already applied is honored on the rescheduled booking.

## 5.5 Payment Status Values

| Status         | Description                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------|
| Pending        | Payment initiated via Razorpay but not yet completed; slot held temporarily                                |
| Successful     | Payment completed; appointment confirmed at discounted total, Priority Service tagged, reminders triggered |
| Failed         | Payment attempt unsuccessful; no appointment created under the pre-payment path; slot released             |
| Expired        | Payment window elapsed without completion; slot released                                                   |
| Not Applicable | Customer chose standard no-payment booking; appointment confirmed without discount or Priority Service tag |

## 5.6 Process Flow

**Step 1** — Customer selects a service and slot, and reaches the booking confirmation step.

**Step 2** — If pre-payment is enabled for the service, the system presents two choices: “Pre-pay now (discount + Priority Service)” or “Book without payment.”

**Step 3** — If the customer chooses standard booking: the appointment is confirmed immediately with no payment, no discount, and no Priority Service tag — proceeds as a normal booking.

**Step 4** — If the customer chooses to pre-pay: the system calculates and displays the discounted total (original price, discount %, discount amount, final payable amount) and holds the slot temporarily.

**Step 5** — Customer is redirected to Razorpay to complete payment.

**Step 6** — Razorpay returns a payment status (Successful or Failed) to SalonBookr.

**Step 7 —** If Successful: appointment is confirmed at the discounted total, the customer is tagged Priority Service, payment record is stored, and the customer receives confirmation and receipt via WhatsApp.

**Step 8 —** Feature 1 reminder scheduling and Feature 2 Priority Service delay handling both begin.

**Step 9 —** If Failed: no appointment is created under the pre-payment path; customer is prompted to retry or switch to standard no-payment booking; the temporarily held slot is released.

**Step 10 —** If the customer abandons the payment flow and the hold window expires, the slot is automatically released for other customers.

## 5.7 Acceptance Criteria

- Given a customer reaches checkout for a pre-payment-enabled service, when the choice is presented, then both “Pre-pay” and “Book without payment” options are available.
- Given a service has a configured discount, when the customer selects pre-pay, then the discounted total is displayed before payment is made.
- Given a customer completes pre-payment successfully, when the booking is confirmed, then the customer is tagged Priority Service and receives the modified Delay Tracker experience.
- Given a pre-payment completes successfully, when the appointment is confirmed, then a payment record (amount, discount, status, transaction ID, timestamp) is stored and the customer receives a WhatsApp receipt.
- Given a customer chooses to pre-pay and payment fails, when the failure is returned by Razorpay, then no appointment record is created and the customer is prompted to retry or switch to standard booking.
- Given a payment is left pending beyond the hold window, when the window expires, then the slot is released back into availability.

## 6. Cross-Feature Business Rules

- Where a Reminder System trigger (Feature 1) and a Delay Tracker trigger (Feature 2) fall at the same point in time — the 2-hour and 1-hour marks — the two messages shall be merged into a single WhatsApp message rather than sent separately, to avoid duplicate notifications to the customer.
- Pre-paid customers (Feature 3) shall be flagged Priority Service in the Scheduling Engine; the Delay Tracker (Feature 2) shall apply a modified, reduced-urgency messaging pattern for these customers unless an actual delay materializes.
- If a Priority Service customer's delay nonetheless exceeds the 60-minute reschedule threshold, the standard Feature 2 reschedule flow applies in full, and any pre-payment discount already applied shall be honored on the rescheduled booking.
- Rescheduling an appointment — whether customer-initiated or delay-triggered — shall not require repayment if pre-payment was already successfully collected for that booking, unless the new slot maps to a different priced service.
- All reminder, delay, and payment events shall be logged with timestamps to support the analytics tracking defined in the BRD.

## 7. Non-Functional Requirements

| Category        | Requirement                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reliability     | Reminder and delay update triggers must fire within $\pm 2$ minutes of their scheduled time                                                            |
| Availability    | WhatsApp and Razorpay integrations should degrade gracefully; failures must be logged, not silently dropped                                            |
| Accuracy        | Delay estimates shown to customers should reflect the salon's actual real-time schedule status within a reasonable margin (target: $\pm 5$ minutes)    |
| Data Integrity  | Payment and appointment state must remain consistent even if Razorpay callback is delayed                                                              |
| Auditability    | Every reminder sent, delay update, and payment transaction must be traceable for analytics and support purposes                                        |
| Configurability | Salon-level settings (30-min reminder, delay cadence, reschedule threshold, discount %, hold windows) must be editable without engineering involvement |

## 8. Out of Scope (Functional)

- SMS or email-based reminders
- Loyalty or rewards logic
- Dynamic or demand-based pricing
- AI-based no-show prediction or demand forecasting

- Support for payment gateways other than Razorpay
- Automated refund processing
- Post-appointment review or feedback collection
- Guaranteed zero-delay SLA for any customer, including Priority Service customers
- Stacking pre-payment discounts with other promotions or loyalty offers
- Mandatory pre-payment requirements of any kind (pre-payment is elective in all cases)

## 9. Open Questions

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- How should the system estimate real-time delay for a salon that does not consistently mark appointments as started or completed?
- Should the discount percentage be a single salon-wide setting, or configurable per service?
- Should partial pre-payments be refundable if a salon cancels the appointment (versus the customer)?
- Should the 24-hour reminder be suppressible for repeat customers with a strong attendance history?
- If a Priority Service customer is offered a reschedule due to delay and selects a different service tier, how is the original discount re-applied?
- What is the fallback communication channel, if any, if a customer's WhatsApp number is invalid or unreachable?